

Chapter 1: Complex Analysis

Mathematical Physics

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Where we are going

To analyze the special functions of physics we first build the tools of *complex analysis*.

- **Taylor and Laurent series** represent a function near a point, even near a singularity.
- **Analytic functions** (complex differentiable) obey the *Cauchy–Riemann equations* and admit power-series expansions.
- **Cauchy’s integral theorem** and the **residue theorem** turn contour integrals into a sum of local residues—the tool used later for Gamma/Beta identities and real integrals.
- A short catalog of **contour templates** (semicircle, keyhole, Jordan) shows how orientation and vanishing-arc estimates convert complex residues into real answers.
- The **method of steepest descent** extracts large-parameter asymptotics of contour integrals. Its real special cases (Laplace method, stationary phase) reappear for Stirling (Ch. 2) and for large-argument Bessel expansions (Ch. 5).

Matching a polynomial to a function

A Taylor polynomial replaces a function near one point by a polynomial with the same value, slope, curvature, and higher derivatives there.

Start at $x = 0$ and write the first few terms rather than a general sum:

$$p(x) = a_0 + a_1x + a_2x^2 + a_3x^3 + \cdots + a_Nx^N \quad (N \geq 3).$$

Differentiate and then set $x = 0$:

$$p(0) = a_0,$$

$$p'(x) = a_1 + 2a_2x + 3a_3x^2 + \cdots,$$

$$p'(0) = a_1,$$

$$p''(x) = 2a_2 + 6a_3x + \cdots,$$

$$p''(0) = 2!a_2,$$

$$p^{(3)}(x) = 6a_3 + \cdots,$$

$$p^{(3)}(0) = 3!a_3.$$

To match f , choose $a_0 = f(0)$, $a_1 = f'(0)$, $a_2 = f''(0)/2!$, and so on. The pattern is

$$a_k = \frac{f^{(k)}(0)}{k!}. \quad (1.1.1)$$

The factorial appears because differentiating x^k exactly k times gives $k!$.

At a different center x_0 , use powers of the distance $x - x_0$:

$$T_N(x; x_0) = \sum_{k=0}^N \frac{f^{(k)}(x_0)}{k!} (x - x_0)^k. \quad (1.1.2)$$

Matching a polynomial to a function (cont.)

Adding more terms need not improve the approximation. The Taylor series equals $f(x)$ when the remainder $R_N(x) = f(x) - T_N(x; x_0)$ tends to zero as $N \rightarrow \infty$:

$$f(x) = \sum_{k=0}^{\infty} \frac{f^{(k)}(x_0)}{k!} (x - x_0)^k. \quad (1.1.3)$$

Having derivatives of every order is not, by itself, enough to guarantee this last equality. The series uses only the numbers $f^{(k)}(x_0)$, and two distinct functions can agree on all of them at a single point. A real example is

$$f(x) = \begin{cases} e^{-1/x^2}, & x \neq 0, \\ 0, & x = 0. \end{cases}$$

For $x \neq 0$ every derivative has the form $p(1/x)e^{-1/x^2}$ with p a polynomial, and each such expression tends to 0 as $x \rightarrow 0$. Hence f has derivatives of every order on $\mathbb{R}$, and all of them vanish at the origin. Its series about the origin is therefore $0 + 0 \cdot x + 0 \cdot x^2 + \dots$, which converges for every x yet reproduces f only at $x = 0$. The series is not failing by diverging; it converges to the wrong function. A smooth f that disagrees with its Taylor series in this way is said to be *not analytic* at that point, and the complex theory below shows that holomorphy rules this out.

Matching a polynomial to a function (cont.)

Exponential. Every derivative of e^x equals e^x , so every derivative at zero equals 1. Therefore

$$e^x = \sum_{k=0}^{\infty} \frac{x^k}{k!} = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots. \quad (1.1.4)$$

The denominator $k!$ grows faster than any power x^k , so the terms shrink rapidly and the series converges for every x , where it represents e^x . For example, at $x = 0.1$ the first three terms give $e^{0.1} \approx 1 + 0.1 + 0.005 = 1.105$.

Cosine and sine. List the derivative values at zero and divide by the corresponding factorial:

k	0	1	2	3	4	5
$(\cos)^{(k)}(0)$	1	0	-1	0	1	0
$(\sin)^{(k)}(0)$	0	1	0	-1	0	1

Only even powers survive for cosine; only odd powers survive for sine:

$$\cos x = \sum_{k=0}^{\infty} \frac{(-1)^k x^{2k}}{(2k)!} = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \cdots, \quad (1.1.5)$$

$$\sin x = \sum_{k=0}^{\infty} \frac{(-1)^k x^{2k+1}}{(2k+1)!} = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \cdots. \quad (1.1.6)$$

Matching a polynomial to a function (cont.)

Every derivative of $\cos x$ and $\sin x$ has magnitude at most 1, so the same factorial dominance makes both series converge for every x . All three series also converge for every complex argument. We use them to extend e^x , $\cos x$, and $\sin x$ to complex numbers.

Imaginary unit, Cartesian and polar form

Define the imaginary unit i by $i^2 = -1$; the other square root of -1 is $-i$. A complex number has Cartesian form

$$z = x + iy, \quad x = \operatorname{Re} z, \quad y = \operatorname{Im} z,$$

where $x, y \in \mathbb{R}$; $\operatorname{Re} z$ and $\operatorname{Im} z$ denote the real and imaginary parts of z . The powers of i repeat with period four:

$$i^{4q} = 1, \quad i^{4q+1} = i, \quad i^{4q+2} = -1, \quad i^{4q+3} = -i.$$

Substitute $i\theta$ into the absolutely convergent exponential series and separate even and odd powers:

$$\begin{aligned} e^{i\theta} &= \sum_{n=0}^{\infty} \frac{(i\theta)^n}{n!} \\ &= \sum_{k=0}^{\infty} \frac{i^{2k} \theta^{2k}}{(2k)!} + \sum_{k=0}^{\infty} \frac{i^{2k+1} \theta^{2k+1}}{(2k+1)!} \end{aligned}$$

and since $i^{2k} = (i^2)^k = (-1)^k$ and $i^{2k+1} = i i^{2k} = i(-1)^k$,

$$= \sum_{k=0}^{\infty} \frac{(-1)^k \theta^{2k}}{(2k)!} + i \sum_{k=0}^{\infty} \frac{(-1)^k \theta^{2k+1}}{(2k+1)!}.$$

Imaginary unit, Cartesian and polar form (cont.)

Comparison with (1.1.5)–(1.1.6) gives Euler's formula

$$e^{i\theta} = \cos \theta + i \sin \theta. \quad (1.2.1)$$

The exponential addition law also holds for complex arguments:

$$e^{w_1+w_2} = e^{w_1} e^{w_2}.$$

It follows by multiplying the two absolutely convergent series and applying the binomial theorem. In particular,

$$e^{u+iv} = e^u (\cos v + i \sin v), \quad u, v \in \mathbb{R}. \quad (1.2.2)$$

The factor e^u determines the modulus, and v determines the angle.

For $z \neq 0$, set

$$r = |z| = \sqrt{x^2 + y^2} > 0.$$

Choose any angle θ satisfying $x = r \cos \theta$ and $y = r \sin \theta$. Then

$$z = r(\cos \theta + i \sin \theta) = re^{i\theta}. \quad (1.2.3)$$

The angle is not unique: if θ is one argument, then every argument is $\theta + 2\pi k$, $k \in \mathbb{Z}$. The principal argument is commonly chosen as $\text{Arg } z \in (-\pi, \pi]$. The number $z = 0$ has modulus 0 but no defined argument.

If $z_1 = r_1 e^{i\theta_1}$ and $z_2 = r_2 e^{i\theta_2}$, then the phases add by the addition law above, so

$$z_1 z_2 = r_1 r_2 e^{i\theta_1} e^{i\theta_2} = r_1 r_2 e^{i(\theta_1 + \theta_2)},$$

Imaginary unit, Cartesian and polar form (cont.)

so moduli multiply and arguments add modulo 2π .

The complex conjugate is

$$\bar{z} = x - iy = re^{-i\theta}, \quad z\bar{z} = (x + iy)(x - iy) = x^2 + y^2 = r^2 = |z|^2,$$

and adding or subtracting z and $\bar{z}$ isolates the Cartesian components:

$$\operatorname{Re} z = \frac{z + \bar{z}}{2}, \quad \operatorname{Im} z = \frac{z - \bar{z}}{2i}. \quad (1.2.4)$$

Replacing θ by $-\theta$ in Euler's formula gives $e^{-i\theta} = \cos \theta - i \sin \theta$. Hence

$$e^{i\theta} + e^{-i\theta} = 2 \cos \theta,$$

$$e^{i\theta} - e^{-i\theta} = 2i \sin \theta,$$

and therefore

$$\cos \theta = \frac{e^{i\theta} + e^{-i\theta}}{2}, \quad \sin \theta = \frac{e^{i\theta} - e^{-i\theta}}{2i}. \quad (1.2.5)$$

Multivalued functions and branch cuts

An inverse relation need not define a single-valued function. For example, $y = x^2$ gives $x = \pm\sqrt{y}$ for $y > 0$. A *branch* selects one value continuously on a chosen domain; the usual real principal square root selects $+\sqrt{y}$ for $y \geq 0$.

The complex logarithm has the same issue: several inputs have the same exponential. **The complex logarithm.** Let $z \neq 0$ and write $z = re^{i\theta}$ with $r > 0$. To solve $e^w = z$, write $w = u + iv$ with $u, v \in \mathbb{R}$. The splitting (1.2.2) gives

$$e^w = e^{u+iv} = e^u(\cos v + i \sin v).$$

Equality with $r(\cos \theta + i \sin \theta)$ requires

$$e^u = r \quad \implies \quad u = \log r,$$

and

$$\cos v = \cos \theta, \quad \sin v = \sin \theta \quad \implies \quad v = \theta + 2\pi k, \quad k \in \mathbb{Z}.$$

Thus the complete multivalued logarithm is

$$\log z = \log |z| + i(\theta + 2\pi k), \quad k \in \mathbb{Z}. \quad (1.2.6)$$

A principal branch is obtained by choosing one interval of length 2π , commonly

$$-\pi < \operatorname{Arg} z < \pi, \quad \operatorname{Log} z = \log |z| + i \operatorname{Arg} z.$$

This analytic branch is defined on $\mathbb{C} \setminus (-\infty, 0]$. The values $\pm\pi$ are limits from the two sides of the cut, not values on this domain.

Multivalued functions and branch cuts (cont.)

A continuous argument increases by 2π after one counterclockwise turn around 0. Thus the logarithm does not return to its starting value: 0 is a *branch point*. Removing a *branch cut* prevents such a loop from staying in our domain. For the principal logarithm, the cut is the nonpositive real axis. Its two boundary values are

$$\operatorname{Log}(-r + i0) = \log r + i\pi, \quad \operatorname{Log}(-r - i0) = \log r - i\pi,$$

so crossing the cut changes the logarithm by $2\pi i$.

The same mechanism applies to roots. For example,

$$z^{1/2} = \sqrt{r} e^{i(\theta+2\pi k)/2} = (-1)^k \sqrt{r} e^{i\theta/2},$$

so there are two values. A branch cut permits one of them to be selected continuously.

The complex derivative and Cauchy–Riemann

The complex derivative at z_0 is

$$f'(z_0) = \lim_{h \rightarrow 0} \frac{f(z_0 + h) - f(z_0)}{h}, \quad (1.3.1)$$

The increment h may approach zero from any direction in the complex plane. The same limit must result in every case. A function is *holomorphic* (analytic) on an open set if this derivative exists at every point there; it is *entire* if that set is all of $\mathbb{C}$.

Write $z = x + iy$ and $f(z) = u(x, y) + iv(x, y)$. Here $u_x = \partial u / \partial x$ means differentiate in x while holding y fixed; u_y , v_x , and v_y are defined in the same way.

Along a real increment $h = \Delta x$, the difference quotient is

$$\begin{aligned} \frac{f(z + h) - f(z)}{h} &= \frac{u(x + \Delta x, y) - u(x, y)}{\Delta x} + i \frac{v(x + \Delta x, y) - v(x, y)}{\Delta x} \\ &\xrightarrow{\Delta x \rightarrow 0} u_x + iv_x, \end{aligned}$$

so $f'(z) = u_x + iv_x$. Along an imaginary increment $h = i\Delta y$ the bookkeeping is the same, with the extra factor $1/i$ pulled out of the denominator:

$$\frac{f(z + i\Delta y) - f(z)}{i\Delta y} = \frac{1}{i} \left[\frac{u(x, y + \Delta y) - u(x, y)}{\Delta y} + i \frac{v(x, y + \Delta y) - v(x, y)}{\Delta y} \right].$$

The complex derivative and Cauchy–Riemann (cont.)

If the complex derivative exists, both directional calculations must give that derivative. Passing to the limit and using $1/i = -i$,

$$f'(z) = \frac{u_y + iv_y}{i} = -i(u_y + iv_y) = v_y - iu_y.$$

Because f is complex differentiable, every direction of approach must return the *same* value $f'(z)$; equating $u_x + iv_x = v_y - iu_y$ and matching real and imaginary parts gives the Cauchy–Riemann equations

$$u_x = v_y, \quad u_y = -v_x. \quad (1.3.2)$$

They are necessary for complex differentiability.

Working test. If u and v have continuous first partial derivatives near a point, the Cauchy–Riemann equations are also sufficient there, and

$$f'(z) = u_x + iv_x = v_y - iu_y.$$

Thus our procedure is: find u, v , compute four partial derivatives, and compare.

Example: z^2 . Since

$$z^2 = (x + iy)^2 = (x^2 - y^2) + i(2xy),$$

we have

$$u_x = 2x = v_y, \quad u_y = -2y = -v_x.$$

Thus z^2 is entire and $f'(z) = 2x + i(2y) = 2z$.

The complex derivative and Cauchy–Riemann (cont.)

Counterexample: $\bar{z} = x - iy$. Then $u_x = 1 \neq -1 = v_y$, so CR fails everywhere. Equivalently $\bar{h}/h$ is 1 for real h and -1 for imaginary h : no direction-independent derivative.

Keep in mind that an analytic function is infinitely differentiable and equals its Taylor series on every disk where it is analytic; we prove this once contour integrals are available.

Contour integrals and a key circle integral

A contour integral is an ordinary integral along a chosen path. Parameterize the path by $z = \gamma(t)$, $a \leq t \leq b$, with γ piecewise continuously differentiable. Since $dz = \gamma'(t) dt$,

$$\int_{\gamma} f(z) dz = \int_a^b f(\gamma(t)) \gamma'(t) dt. \quad (1.4.1)$$

Reversing the path reverses the sign:

$$\int_{-\gamma} f(z) dz = - \int_{\gamma} f(z) dz.$$

A closed contour may be traversed in either direction. A positively oriented boundary keeps the region on the left: the outer boundary is counterclockwise, while a boundary around a hole is clockwise.

Integer powers on a circle. Let $n \in \mathbb{Z}$ and parameterize the positively oriented circle $|z| = R$ by

$$z(\theta) = Re^{i\theta}, \quad dz = iRe^{i\theta} d\theta, \quad 0 \leq \theta \leq 2\pi.$$

Contour integrals and a key circle integral (cont.)

Then

$$\begin{aligned}\oint_{|z|=R} z^n dz &= \int_0^{2\pi} R^n e^{in\theta} (iRe^{i\theta}) d\theta \\ &= iR^{n+1} \int_0^{2\pi} e^{i(n+1)\theta} d\theta.\end{aligned}$$

If $n = -1$, the exponential is 1, so

$$\oint_{|z|=R} \frac{dz}{z} = i \int_0^{2\pi} d\theta = 2\pi i.$$

If $n \neq -1$, then

$$\begin{aligned}iR^{n+1} \int_0^{2\pi} e^{i(n+1)\theta} d\theta &= iR^{n+1} \left[\frac{e^{i(n+1)\theta}}{i(n+1)} \right]_0^{2\pi} \\ &= \frac{R^{n+1}}{n+1} (e^{2\pi i(n+1)} - 1) = 0,\end{aligned}$$

because $n+1$ is an integer. Let $\delta_{n,-1}$ denote the Kronecker delta: it is 1 when $n = -1$ and 0 otherwise. Translating the circle gives

$$\oint_{|z-z_0|=R} (z-z_0)^n dz = 2\pi i \delta_{n,-1}. \quad (1.4.2)$$

Cauchy's integral theorem and formula

A *domain* is an open, connected set. A domain is *simply connected* if every closed curve in it can be continuously contracted to a point without leaving the domain; for the planar regions used here, this is the precise version of “the region has no holes.”

Cauchy's integral theorem

Let D be a simply connected domain, let f be holomorphic on D , and let γ be a closed piecewise smooth contour in D . Then

$$\oint_{\gamma} f(z) dz = 0.$$

Cauchy's integral theorem and formula (cont.)

Why zero? First let $\gamma = \partial E$ be a positively oriented simple closed contour. Green's theorem and the Cauchy–Riemann equations give, with $f = u + iv$ and $dz = dx + i dy$,

$$\begin{aligned}\oint_{\gamma} f dz &= \oint_{\gamma} (u dx - v dy) + i \oint_{\gamma} (v dx + u dy) \\ &= \iint_E (-v_x - u_y) dA + i \iint_E (u_x - v_y) dA \\ &= \iint_E (u_y - u_y) dA + i \iint_E (v_y - v_y) dA = 0.\end{aligned}$$

This elementary calculation assumes that u and v have continuous first partial derivatives near E . The full Cauchy–Goursat theorem removes that extra regularity assumption and gives the stated conclusion for every piecewise smooth closed contour in a simply connected domain; holomorphy alone is enough.

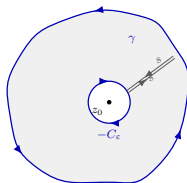
Cauchy's integral theorem and formula (cont.)

Cauchy's integral formula

Let γ be a positively oriented simple closed contour, let f be holomorphic on an open set containing γ and its interior, and let z_0 lie inside γ . Then

$$f(z_0) = \frac{1}{2\pi i} \oint_{\gamma} \frac{f(z)}{z - z_0} dz. \quad (1.4.3)$$

To understand the formula, deform γ to a small counterclockwise circle C_ε about z_0 . The integrand is holomorphic between the two curves, so their integrals agree. The two sides of the connecting slit cancel.



Slit construction: the combined contour $\gamma + s - C_\varepsilon - s$ encloses the shaded region, where the integrand is holomorphic; the inner circle is traversed clockwise and the slit sides s and $-s$ cancel.

Cauchy's integral theorem and formula (cont.)

On the small circle $z = z_0 + \varepsilon e^{i\theta}$,

$$\oint_{C_\varepsilon} \frac{f(z)}{z - z_0} dz = i \int_0^{2\pi} f(z_0 + \varepsilon e^{i\theta}) d\theta.$$

As $\varepsilon \rightarrow 0$, continuity makes the integrand approach $f(z_0)$, so the last integral approaches $2\pi i f(z_0)$. This gives (1.4.3).

Differentiate (1.4.3) in z_0 (contour fixed, away from z_0). Since $\partial_{z_0}^n (z - z_0)^{-1} = n! / (z - z_0)^{n+1}$, for $n = 0, 1, 2, \dots$,

$$f^{(n)}(z_0) = \frac{n!}{2\pi i} \oint_\gamma \frac{f(z)}{(z - z_0)^{n+1}} dz. \quad (1.4.4)$$

In particular every holomorphic function is infinitely differentiable, and for $m = 1, 2, \dots$,

$$\oint_\gamma \frac{f(z)}{(z - z_0)^m} dz = \frac{2\pi i}{(m-1)!} f^{(m-1)}(z_0).$$

Taylor series from contour integration

Taylor series theorem

If f is holomorphic on the disk $|z - z_0| < R$, then for every $|z - z_0| < R$,

$$f(z) = \sum_{n=0}^{\infty} \frac{f^{(n)}(z_0)}{n!} (z - z_0)^n.$$

Choose a radius ρ satisfying

$$|z - z_0| < \rho < R$$

and integrate on $C_\rho : |\xi - z_0| = \rho$. Cauchy's formula gives

$$f(z) = \frac{1}{2\pi i} \oint_{C_\rho} \frac{f(\xi)}{\xi - z} d\xi.$$

The idea is to expand only the kernel, using the geometric series:

$$\frac{1}{\xi - z} = \frac{1}{\xi - z_0} \frac{1}{1 - (z - z_0)/(\xi - z_0)} = \sum_{n=0}^{\infty} \frac{(z - z_0)^n}{(\xi - z_0)^{n+1}}.$$

Taylor series from contour integration (cont.)

This works because $|(z - z_0)/(\xi - z_0)| = |z - z_0|/\rho < 1$ all along the circle. The geometric series therefore converges uniformly on the fixed circle, so we may substitute it and integrate term by term:

$$f(z) = \sum_{n=0}^{\infty} (z - z_0)^n \underbrace{\frac{1}{2\pi i} \oint_{C_\rho} \frac{f(\xi)}{(\xi - z_0)^{n+1}} d\xi}_{f^{(n)}(z_0)/n! \text{ by (1.4.4)}}.$$

The coefficients are exactly those found by matching derivatives at the start of the chapter. Here Cauchy's formula also tells us where the series equals f .

The key consequence is simple: inside a disk where f is holomorphic, its Taylor series represents f . For an expansion about z_0 , the nearest nonremovable singularity limits the radius of convergence. Boundary points require separate checking.

The converse also holds: a convergent power series may be differentiated term by term inside its disk, so it is holomorphic there. Being analytic at z_0 and having a Taylor series representation about z_0 are the same property.

Analytic continuation

A Taylor-series expansion is local: it may stop converging even though the function itself remains holomorphic farther away.

For example,

$$f(z) = \frac{1}{1-z} = 1 + z + z^2 + \cdots, \quad |z| < 1.$$

The Taylor series about $z = 0$ converges only for $|z| < 1$, because the nearest singularity is at $z = 1$. However, the function

$$f(z) = \frac{1}{1-z}$$

is still holomorphic at every point except $z = 1$.

To go beyond the original Taylor disk, choose a new center inside the region where f is already known. For example, about $z = -\frac{1}{2}$,

$$\begin{aligned} f(z) &= \frac{1}{\frac{3}{2} - (z + \frac{1}{2})} \\ &= \frac{2}{3} \frac{1}{1 - \frac{2}{3}(z + \frac{1}{2})} \\ &= \frac{2}{3} \sum_{n=0}^{\infty} \left[\frac{2}{3} \left(z + \frac{1}{2} \right) \right]^n, \end{aligned}$$

Analytic continuation (cont.)

which converges for

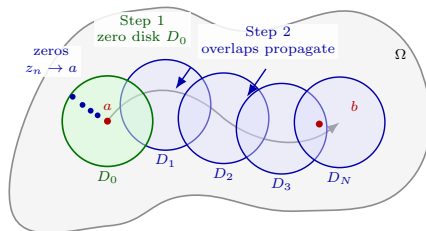
$$\left| z + \frac{1}{2} \right| < \frac{3}{2}.$$

This new Taylor disk overlaps the original disk but reaches farther.

More generally, we may repeat this process:

$$D_0 \longrightarrow D_1 \longrightarrow D_2 \longrightarrow \cdots .$$

On every overlap, the Taylor expansions represent the same holomorphic function and therefore must agree. In this way, the function can be continued step by step through a larger connected domain.



Laurent series

A Taylor series uses nonnegative powers. Near a singularity we also need $1/(z - z_0), 1/(z - z_0)^2, \dots$. Their combination is a *Laurent series*.

Suppose f is holomorphic in the annulus

$$r < |z - z_0| < R.$$

An inner hole allows negative powers; the next singularity limits the outer radius. f has a Laurent expansion

$$f(z) = \sum_{n=-\infty}^{\infty} a_n (z - z_0)^n = \underbrace{\sum_{n=0}^{\infty} a_n (z - z_0)^n}_{\text{regular part}} + \underbrace{\sum_{n=1}^{\infty} a_{-n} (z - z_0)^{-n}}_{\text{principal part}}, \quad (1.5.1)$$

convergent throughout the annulus. To extract a coefficient, multiply by $(z - z_0)^{-k-1}$ and integrate on a centered circle $\gamma : |z - z_0| = \rho$, where $r < \rho < R$, oriented counterclockwise:

$$\begin{aligned} \frac{1}{2\pi i} \oint_{\gamma} \frac{f(z)}{(z - z_0)^{k+1}} dz &= \frac{1}{2\pi i} \sum_{n=-\infty}^{\infty} a_n \oint_{\gamma} (z - z_0)^{n-k-1} dz \\ &= \frac{1}{2\pi i} a_k (2\pi i) = a_k. \end{aligned}$$

Laurent series (cont.)

Therefore, for every integer k ,

$$a_k = \frac{1}{2\pi i} \oint_{\gamma} \frac{f(z)}{(z - z_0)^{k+1}} dz. \quad (1.5.2)$$

On the chosen circle inside the annulus, term-by-term integration is valid.

If the principal part is empty, the series extends holomorphically to the full disk and is a Taylor series.

Classifying singularities

A point z_0 is a *singular point* of f when f fails to be holomorphic there. It is an *isolated singularity* if there exists $R > 0$ such that f is holomorphic on the punctured disk

$$0 < |z - z_0| < R.$$

Only isolated singularities admit an ordinary Laurent classification. Write

$$f(z) = \sum_{n=-\infty}^{\infty} a_n(z - z_0)^n.$$

Then:

- The singularity is **removable** if $a_n = 0$ for every $n < 0$. Defining $f(z_0) = a_0$ makes f holomorphic at z_0 . Example: $\sin z/z$ at 0.
- It is a **pole of order m** , where $m \geq 1$ is an integer, if $a_{-m} \neq 0$ and $a_n = 0$ for every $n < -m$. Equivalently,

$$f(z) = \frac{h(z)}{(z - z_0)^m}, \quad h \text{ holomorphic, } h(z_0) \neq 0.$$

Multiplying by $(z - z_0)^m$ removes exactly the highest negative power.

Classifying singularities (cont.)

- It is **essential** if infinitely many negative-power coefficients are nonzero. For example,

$$\sin\left(\frac{1}{z}\right) = \frac{1}{z} - \frac{1}{3!z^3} + \frac{1}{5!z^5} - \dots$$

has an essential singularity at 0.

A function holomorphic except for isolated poles is *meromorphic*. For example, $1/\sin z$ is meromorphic, with simple poles at $z = n\pi$, $n \in \mathbb{Z}$.

A branch point is not an isolated singularity. For example, analytic continuation of $\log z$ once around 0 changes its value by $2\pi i$, so no single-valued branch is holomorphic on a complete punctured disk around 0.

At infinity, inspect $F(w) = f(1/w)$ at $w = 0$. A polynomial of degree $m \geq 1$ has a pole of order m there.

The residue theorem

For an isolated singularity z_0 , the residue is the Laurent coefficient

$$\operatorname{Res}(f, z_0) = a_{-1}.$$

Choose a small counterclockwise circle C about z_0 enclosing no other singularity. Use the Laurent expansion valid near z_0 , not one on an annulus containing other singularities. By (1.4.2), integrating any integer power around C gives zero except for $(z - z_0)^{-1}$, whose integral is $2\pi i$. Only the $n = -1$ term survives:

$$\begin{aligned}\oint_C f(z) dz &= \sum_{n=-\infty}^{\infty} a_n \oint_C (z - z_0)^n dz \\ &= a_{-1}(2\pi i) \\ &= 2\pi i \operatorname{Res}(f, z_0).\end{aligned}$$

Thus

$$\oint_C f(z) dz = 2\pi i \operatorname{Res}(f, z_0). \quad (1.7.1)$$

The residue theorem (cont.)

Residue theorem

Let f be holomorphic inside and on a positively oriented simple closed contour γ , except for finitely many isolated singularities $z_1, \dots, z_N$ in the interior. Then

$$\oint_{\gamma} f(z) dz = 2\pi i \sum_{k=1}^N \operatorname{Res}(f, z_k). \quad (1.7.2)$$

Geometric reason. Deform the outer contour into small counterclockwise circles around the singularities. Each circle contributes its own $2\pi i$ times residue. The simple outer contour encloses each of these singularities once.

Residue at a pole of order m . Write

$$h(z) = (z - z_0)^m f(z),$$

where h is holomorphic and $h(z_0) \neq 0$. Taylor expansion gives

$$h(z) = \sum_{j=0}^{\infty} \frac{h^{(j)}(z_0)}{j!} (z - z_0)^j.$$

The residue theorem (cont.)

Dividing by $(z - z_0)^m$,

$$f(z) = \sum_{j=0}^{\infty} \frac{h^{(j)}(z_0)}{j!} (z - z_0)^{j-m}.$$

The exponent equals -1 exactly when $j = m - 1$. Therefore

$$\operatorname{Res}(f, z_0) = \frac{1}{(m-1)!} \left. \frac{d^{m-1}}{dz^{m-1}} [(z - z_0)^m f(z)] \right|_{z=z_0}. \quad (1.7.3)$$

For $m = 1$ this reduces to

$$\operatorname{Res}(f, z_0) = \lim_{z \rightarrow z_0} (z - z_0) f(z),$$

and for $m = 2$ it becomes the first derivative of $(z - z_0)^2 f(z)$ at z_0 .

A quotient at a simple zero. If $q(z_0) = 0$ and $q'(z_0) \neq 0$, then

$$q(z) = q'(z_0)(z - z_0) + \cdots.$$

For p, q holomorphic near z_0 , cancel $z - z_0$ before taking the limit:

$$\operatorname{Res}\left(\frac{p}{q}, z_0\right) = \lim_{z \rightarrow z_0} \frac{p(z)}{q(z)/(z - z_0)} = \frac{p(z_0)}{q'(z_0)}. \quad (1.7.4)$$

The residue theorem (cont.)

When $p(z_0) = 0$, the singularity is removable and the formula gives zero.
For example,

$$\operatorname{Res}\left(\frac{1}{z^2+1}, i\right) = \frac{1}{2i},$$

while the order-2 formula at the double pole of $1/(z^2+1)^2$ gives

$$\operatorname{Res}\left(\frac{1}{(z^2+1)^2}, i\right) = \left.\frac{d}{dz} \frac{1}{(z+i)^2}\right|_{z=i} = -\frac{2}{(2i)^3} = \frac{1}{4i}.$$

Quick checks on $|z| = 1$.

$$\frac{\sin z}{z^2} = \frac{1}{z} - \frac{z}{3!} + \cdots \implies \operatorname{Res}\left(\frac{\sin z}{z^2}, 0\right) = 1,$$

so $\oint_{|z|=1} (\sin z)/z^2 dz = 2\pi i$. By contrast $z^2/\sin z$ extends holomorphically through 0 (value 0) and has no other zeros of $\sin z$ inside the unit disk, so

$$\oint_{|z|=1} \frac{z^2}{\sin z} dz = 0.$$

Real integrals by a semicircular contour

Let Γ_R consist of the real segment from $-R$ to R and the upper semicircle $C_R : z = Re^{i\theta}$, $0 \leq \theta \leq \pi$. For an integrand with no pole on the real segment,

$$\oint_{\Gamma_R} f(z) dz = \int_{-R}^R f(x) dx + \int_{C_R} f(z) dz. \quad (1.8.1)$$

The magnitude of an integral is at most the maximum integrand magnitude M times the path length L ; this is the ML estimate. If $M_R = \max_{z \in C_R} |f(z)|$, the arc length is πR , so

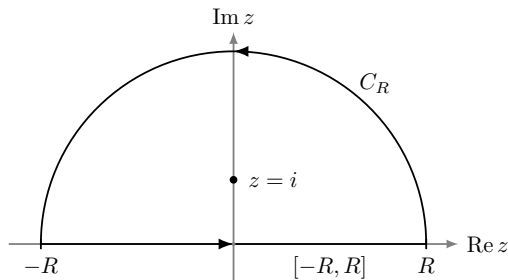
$$\left| \int_{C_R} f(z) dz \right| \leq \pi R M_R.$$

Thus the arc vanishes whenever $RM_R \rightarrow 0$. If the improper real integral converges, the residue theorem then gives

$$\int_{-\infty}^{\infty} f(x) dx = 2\pi i \sum_{\operatorname{Im} z_k > 0} \operatorname{Res}(f, z_k).$$

When an exponential e^{iaz} is present, its decay in one half-plane often supplies the needed estimate.

Real integrals by a semicircular contour (cont.)



Example: $\int_{-\infty}^{\infty} \cos x / (1 + x^2) dx$. Consider

$$J = \int_{-\infty}^{\infty} \frac{e^{ix}}{1 + x^2} dx.$$

For $z = x + iy$ in the upper half-plane,

$$|e^{iz}| = |e^{ix-y}| = e^{-y} \leq 1.$$

On $|z| = R > 1$,

$$|1 + z^2| \geq ||z|^2 - 1| = R^2 - 1,$$

Real integrals by a semicircular contour (cont.)

so

$$\left| \int_{C_R} \frac{e^{iz}}{1+z^2} dz \right| \leq \frac{\pi R}{R^2-1} \rightarrow 0.$$

Only the pole $z = i$ lies in the upper half-plane. Since $1+z^2 = (z-i)(z+i)$,

$$\begin{aligned} \operatorname{Res} \left(\frac{e^{iz}}{1+z^2}, i \right) &= \frac{e^{iz}}{z+i} \Big|_{z=i} \\ &= \frac{e^{-1}}{2i}. \end{aligned}$$

Hence

$$J = 2\pi i \frac{e^{-1}}{2i} = \frac{\pi}{e}.$$

Taking real parts gives

$$\int_{-\infty}^{\infty} \frac{\cos x}{1+x^2} dx = \frac{\pi}{e}. \quad (1.8.2)$$

The imaginary part vanishes consistently because $\sin x/(1+x^2)$ is odd.

Real integrals by a semicircular contour (cont.)

Example: $\int_{-\infty}^{\infty} dx/(1+x^4)$. Close in the upper half-plane. The upper arc vanishes by ML: $|\int_{C_R}| \leq \pi R/(R^4-1) \rightarrow 0$ as $R \rightarrow \infty$. The poles are the fourth roots of -1 :

$$z_k = e^{i(\pi+2k\pi)/4}, \quad k = 0, 1, 2, 3.$$

Those in the upper half-plane are the simple poles

$$z_0 = e^{i\pi/4}, \quad z_1 = e^{3\pi/4}.$$

Since $1+z^4 = \prod_{k=0}^3 (z-z_k)$ and $(1+z^4)' = 4z^3$, each residue is

$$\text{Res}\left(\frac{1}{1+z^4}, z_j\right) = \frac{1}{4z_j^3}.$$

Adding the two upper-half-plane residues,

$$\sum_{j=0}^1 \text{Res}\left(\frac{1}{1+z^4}, z_j\right) = -\frac{e^{i\pi/4} + e^{3i\pi/4}}{4} = -\frac{i}{2\sqrt{2}}.$$

The residue theorem therefore gives

$$\int_{-\infty}^{\infty} \frac{dx}{1+x^4} = 2\pi i \left(-\frac{i}{2\sqrt{2}}\right) = \frac{\pi}{\sqrt{2}}. \quad (1.8.3)$$

Keyhole contour: $\int_0^\infty x^a/(1+x)^2 dx$

Let

$$I(a) = \int_0^\infty \frac{x^a}{(1+x)^2} dx, \quad -1 < a < 1.$$

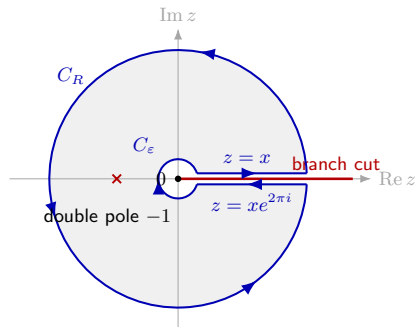
The endpoint conditions follow directly: near 0 the integrand behaves as x^a , which is integrable when $a > -1$; near infinity it behaves as x^{a-2} , which is integrable when $a < 1$.

For $a \neq 0$, choose the branch of the power whose cut runs along the *positive* real axis,

$$z^a = e^{a\ell(z)}, \quad \ell(z) = \log |z| + i \arg z, \quad 0 < \arg z < 2\pi,$$

rather than the principal logarithm used earlier. The cut is chosen along the real integration interval so that the keyhole follows its two sides. Use a counterclockwise keyhole with outer radius R and inner radius ε . The hole keeps the branch point 0 outside the enclosed region. A branch point has no ordinary residue: the power is not single-valued on a full punctured disk.

Keyhole contour: $\int_0^\infty x^a/(1+x)^2 dx$ (cont.)



On the upper side, $z = x$ and the contribution is

$$I_{\varepsilon,R}(a) = \int_{\varepsilon}^R \frac{x^a}{(1+x)^2} dx.$$

On the lower side, $z = x e^{2\pi i}$ and $z^a = x^a e^{2\pi i a}$. The orientation is from R back to ε , so the contribution is

$$\int_R^{\varepsilon} \frac{x^a e^{2\pi i a}}{(1+x)^2} dx = -e^{2\pi i a} I_{\varepsilon,R}(a).$$

Keyhole contour: $\int_0^\infty x^a/(1+x)^2 dx$ (cont.)

Thus the two straight segments sum to

$$(1 - e^{2\pi ia})I_{\varepsilon,R}(a).$$

Both arcs die by the *ML* estimate. On $|z| = R > 1$, $|z^a| = R^a$ and $|1+z| \geq R-1$, while the arc length is $2\pi R$. Hence

$$\left| \int_{C_R} \frac{z^a}{(1+z)^2} dz \right| \leq \frac{2\pi R^{a+1}}{(R-1)^2} = O(R^{a-1}) \rightarrow 0$$

because $a < 1$. On $|z| = \varepsilon < 1$, $|z^a| = \varepsilon^a$, $|1+z| \geq 1-\varepsilon$, and the arc length is $2\pi\varepsilon$, so

$$\left| \int_{C_\varepsilon} \frac{z^a}{(1+z)^2} dz \right| \leq \frac{2\pi\varepsilon^{a+1}}{(1-\varepsilon)^2} = O(\varepsilon^{a+1}) \rightarrow 0$$

because $a > -1$. Here $F = O(G)$ means $|F|$ is bounded by a constant times $|G|$ in the stated limit.

The only enclosed singularity is the double pole at $z = -1$. Since

$$(z+1)^2 \frac{z^a}{(1+z)^2} = z^a,$$

Keyhole contour: $\int_0^\infty x^a/(1+x)^2 dx$ (cont.)

formula (1.7.3) gives

$$\operatorname{Res}\left(\frac{z^a}{(1+z)^2}, -1\right) = \left.\frac{d}{dz}z^a\right|_{z=-1}.$$

The chain rule gives $(z^a)' = az^{a-1}$ on this branch. Since $\ell(-1) = i\pi$,

$$\operatorname{Res}\left(\frac{z^a}{(1+z)^2}, -1\right) = ae^{i\pi(a-1)} = -ae^{i\pi a}.$$

Hence the residue theorem gives, after $R \rightarrow \infty$ and $\varepsilon \rightarrow 0$,

$$(1 - e^{2\pi ia})I(a) = -2\pi iae^{i\pi a}.$$

Factor the denominator:

$$\begin{aligned}1 - e^{2\pi ia} &= e^{i\pi a}(e^{-i\pi a} - e^{i\pi a}) \\&= -2ie^{i\pi a}\sin(\pi a).\end{aligned}$$

Therefore

$$\begin{aligned}I(a) &= \frac{-2\pi iae^{i\pi a}}{-2ie^{i\pi a}\sin(\pi a)} \\&= \frac{\pi a}{\sin(\pi a)}.\end{aligned}$$

Keyhole contour: $\int_0^\infty x^a/(1+x)^2 dx$ (cont.)

Thus

$$\int_0^\infty \frac{x^a}{(1+x)^2} dx = \frac{\pi a}{\sin(\pi a)}, \quad -1 < a < 1. \quad (1.8.4)$$

At $a = 0$, the right-hand side has limit 1 because $\sin(\pi a) \sim \pi a$, where $u(a) \sim v(a)$ means $u(a)/v(a) \rightarrow 1$ in the stated limit, and directly $\int_0^\infty (1+x)^{-2} dx = 1$.

The same keyhole with the simpler integrand $z^{-a}/(1+z)$ (simple pole at -1) yields the companion evaluation, valid for $0 < a < 1$,

$$\int_0^\infty \frac{x^{-a}}{1+x} dx = \frac{\pi}{\sin(\pi a)}. \quad (1.8.5)$$

Here the lower-side phase is $e^{-2\pi ia}$, and the residue at -1 is $e^{-i\pi a}$. The outer and inner arcs are respectively $O(R^{-a})$ and $O(\varepsilon^{1-a})$, so both vanish. Therefore

$$(1 - e^{-2\pi ia}) \int_0^\infty \frac{x^{-a}}{1+x} dx = 2\pi i e^{-i\pi a}.$$

Using $1 - e^{-2\pi ia} = 2ie^{-i\pi a} \sin(\pi a)$ gives (1.8.5). This reflection factor reappears in the Gamma and Beta functions.

Jordan's lemma and $\int \sin x/x \, dx$

A small clockwise semicircle around a real-axis pole. Near a simple pole z_0 , write

$$f(z) = \frac{r_{-1}}{z - z_0} + g(z), \quad r_{-1} = \operatorname{Res}(f, z_0),$$

with g bounded. Parameterize the upper indentation by $z = z_0 + \varepsilon e^{i\theta}$, traversed from $\theta = \pi$ to 0. Then

$$\frac{r_{-1}}{z - z_0} dz = ir_{-1} d\theta, \quad ir_{-1} \int_{\pi}^0 d\theta = -i\pi r_{-1}.$$

The bounded part contributes at most its bound times $\pi\varepsilon$, so

$$\int_{C_{\varepsilon}} f(z) dz \longrightarrow -i\pi \operatorname{Res}(f, z_0). \quad (1.8.6)$$

The minus sign comes from the clockwise orientation.

Jordan's lemma and $\int \sin x/x \, dx$ (cont.)

Jordan's lemma

Suppose that f is holomorphic in the upper half-plane except at finitely many isolated singularities and that $|f(z)| \rightarrow 0$ as $|z| \rightarrow \infty$ there. Let C_R be the upper semicircle, chosen to avoid those singularities. Then, for every $a > 0$,

$$\int_{C_R} f(z) e^{iaz} \, dz \rightarrow 0.$$

Proof. Parameterize C_R by $z = Re^{i\theta}$, $0 \leq \theta \leq \pi$, and set

$$M_R = \max_{z \in C_R} |f(z)|.$$

Because

$$e^{iaz} = e^{iaR \cos \theta} e^{-aR \sin \theta}, \quad |dz| = R \, d\theta,$$

we obtain

$$\begin{aligned} \left| \int_{C_R} f(z) e^{iaz} \, dz \right| &\leq M_R R \int_0^\pi e^{-aR \sin \theta} \, d\theta \\ &= 2M_R R \int_0^{\pi/2} e^{-aR \sin \theta} \, d\theta. \end{aligned}$$

Jordan's lemma and $\int \sin x/x \, dx$ (cont.)

On $0 \leq \theta \leq \pi/2$, concavity of sine gives $\sin \theta \geq 2\theta/\pi$. Therefore

$$\begin{aligned} \left| \int_{C_R} f(z) e^{iaz} \, dz \right| &\leq 2M_R R \int_0^{\pi/2} e^{-2aR\theta/\pi} \, d\theta \\ &= \frac{\pi M_R}{a} (1 - e^{-aR}) \\ &\leq \frac{\pi M_R}{a} \longrightarrow 0. \end{aligned}$$

The last limit follows from the stated decay, which gives $M_R \rightarrow 0$. This bound also controls the small angular regions near $\theta = 0$ and π , where the exponential alone does not decay. Analyticity is used when the arc is joined to a contour and the residue theorem is applied; the bound itself uses only the decay of M_R .

Example: the Dirichlet integral. Integrate e^{iz}/z over the upper semicircle, indenting clockwise above $z = 0$. On the large arc, take $f(z) = 1/z$, so $M_R = 1/R \rightarrow 0$ and the Jordan estimate applies. The small semicircle over $z = 0$ is

Jordan's lemma and $\int \sin x/x \, dx$ (cont.)

traversed clockwise through angle π , so by the indentation rule (1.8.6) it contributes $-i\pi \operatorname{Res}$:

$$\begin{aligned} -i\pi \operatorname{Res} \left(\frac{e^{iz}}{z}, 0 \right) &= -i\pi \lim_{z \rightarrow 0} e^{iz} \\ &= -i\pi. \end{aligned}$$

The indentation excludes the only pole, so e^{iz}/z is holomorphic on and inside the indented contour and Cauchy's theorem makes the total integral zero. Written out, that contour has four pieces — the segment $[-R, -\varepsilon]$, the small semicircle C_ε traversed clockwise from $-\varepsilon$ to $+\varepsilon$, the segment $[\varepsilon, R]$, and the large semicircle C_R :

$$\int_{-R}^{-\varepsilon} \frac{e^{ix}}{x} dx + \int_{C_\varepsilon} \frac{e^{iz}}{z} dz + \int_{\varepsilon}^R \frac{e^{ix}}{x} dx + \int_{C_R} \frac{e^{iz}}{z} dz = 0.$$

The two real segments must be combined symmetrically because $\cos x/x$ behaves as $1/x$ near zero. This is the *Cauchy principal value*:

$$\operatorname{PV} \int_{-\infty}^{\infty} g(x) dx = \lim_{\substack{\varepsilon \rightarrow 0^+ \\ R \rightarrow \infty}} \left(\int_{-R}^{-\varepsilon} g(x) dx + \int_{\varepsilon}^R g(x) dx \right).$$

Jordan's lemma and $\int \sin x/x \, dx$ (cont.)

The large arc tends to 0 and the small arc to $-i\pi$. Letting $R \rightarrow \infty$ and $\varepsilon \rightarrow 0$ in the four-piece identity therefore gives

$$0 = \text{PV} \int_{-\infty}^{\infty} \frac{e^{ix}}{x} \, dx - i\pi,$$

and hence

$$\text{PV} \int_{-\infty}^{\infty} \frac{e^{ix}}{x} \, dx = i\pi. \quad (1.8.7)$$

Taking imaginary parts,

$$\int_{-\infty}^{\infty} \frac{\sin x}{x} \, dx = \pi. \quad (1.8.8)$$

The integrand $\sin x/x$ is continuous at 0 after assigning the value 1, so its integral needs no principal-value interpretation there; the principal value was required only for the auxiliary exponential integral.

Sampling kernels: writing a sum as an integral

Residue calculus also runs in the other direction: a *sum* can be written as a contour integral by multiplying the summand by a kernel whose poles sit at the sample points.

Integer samples. The kernel $\pi \cot \pi z$ has a simple pole at every integer n with residue

$$\operatorname{Res}_{z=n} \pi \cot \pi z = \frac{\pi \cos \pi n}{\pi \cos \pi n} = 1.$$

For f holomorphic on and inside a positively oriented contour C that avoids the integers and encloses $n_1, \dots, n_2$ (and no other poles of the integrand),

$$\sum_{m=n_1}^{n_2} f(m) = \frac{1}{2\pi i} \oint_C f(z) \pi \cot \pi z \, dz.$$

The alternating variant uses $\pi \csc \pi z = \pi / \sin \pi z$, whose residue at n is $(-1)^n$, giving $\sum (-1)^m f(m)$.

Half-integer samples. The kernel $\pi / \cos \pi z$ has residue $(-1)^{n+1}$ at $z = n + \frac{1}{2}$ (since $(\cos \pi z)'|_{n+1/2} = -\pi(-1)^n$), so a contour enclosing the half-integers samples $\sum (-1)^{n+1} f(n + \frac{1}{2})$.

The payoff comes after the rewrite: deforming C moves the value of the sum onto the poles and saddle contributions of f itself. Turning a slowly converging modal

Sampling kernels: writing a sum as an integral (cont.)

series into residues at complex poles of the summand is the Watson transformation of scattering theory; this frame is its entire complex analysis.

The steepest-descent approximation

Step 1: set up the large-parameter integral. Many asymptotic formulas for special functions come from integrals of the form

$$I(s) = \int_{\Gamma} g(z) e^{sf(z)} dz, \quad s \rightarrow +\infty, \quad (1.9.1)$$

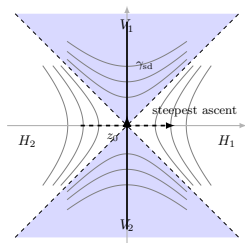
where f and g are holomorphic near a contour Γ that may be deformed without crossing singularities of the integrand. Write $f = u + iv$ with real u, v . As s grows, e^{su} favors points where u is largest on the deformed contour. We study the case where a *saddle* supplies the leading contribution; endpoints must not dominate.

Step 2: locate a nondegenerate saddle. We seek a point z_0 where

$$f'(z_0) = 0, \quad f''(z_0) \neq 0.$$

At such a point the first-order change vanishes. The quadratic term determines which directions make $\operatorname{Re} f$ decrease. In the complex plane these are descent directions through a saddle, rather than a maximum in every direction.

The steepest-descent approximation (cont.)



In the schematic, hill sectors and valley sectors meet at the saddle; a steepest-descent path through the saddle runs into the valleys of $u = \operatorname{Re} f$.

Step 3: choose the descent direction. Taylor expansion at a nondegenerate saddle is

$$f(z) = f(z_0) + \frac{1}{2} f''(z_0)(z - z_0)^2 + O((z - z_0)^3). \quad (1.9.2)$$

Write

$$f''(z_0) = |f''(z_0)| e^{i\theta}, \quad z - z_0 = r e^{i\phi}.$$

Then

$$\frac{1}{2} f''(z_0)(z - z_0)^2 = \frac{1}{2} |f''(z_0)| r^2 e^{i(\theta+2\phi)}.$$

The steepest-descent approximation (cont.)

A *steepest-descent* direction makes this quadratic change negative real, so

$$\theta + 2\phi = \pi \pmod{2\pi}, \quad \text{that is} \quad \phi = \frac{\pi - \theta}{2} \pmod{\pi}. \quad (1.9.3)$$

The ambiguity by π is not a defect: it is the two opposite rays leaving z_0 , and Step 4 uses both at once by letting r run over all of $\mathbb{R}$. Along that ray,

$$\frac{1}{2}f''(z_0)(z - z_0)^2 = -\frac{1}{2}|f''(z_0)|r^2.$$

(The orthogonal choice $\theta + 2\phi = 0$ is *steepest ascent*.)

Step 4: reduce to a Gaussian integral. Assume the contour can be deformed through z_0 along both descent rays, that this saddle dominates the integral, and that $g(z_0) \neq 0$. Parameterize the full descent line by

$$\alpha = e^{i(\pi-\theta)/2} = i e^{-i\theta/2}, \quad z = z_0 + \alpha r, \quad r \in \mathbb{R},$$

Thus $dz = \alpha dr$. Since $\alpha^2 = -e^{-i\theta}$, we have $f''(z_0)\alpha^2 = -|f''(z_0)|$. Replacing f by its quadratic Taylor piece and g by the constant $g(z_0)$ yields

$$\begin{aligned} I(s) &\sim g(z_0)e^{sf(z_0)}\alpha \int_{-\infty}^{\infty} \exp\left(-\frac{1}{2}s|f''(z_0)|r^2\right) dr \\ &= g(z_0)e^{sf(z_0)}\alpha \sqrt{\frac{2\pi}{s|f''(z_0)|}}, \end{aligned}$$

The steepest-descent approximation (cont.)

where we used $\int_{-\infty}^{\infty} e^{-ar^2} dr = \sqrt{\pi/a}$ with $a = \frac{1}{2}s|f''(z_0)|$. This standard Gaussian integral is evaluated in Chapter 2.

Reverse orientation changes the sign; one descent ray gives half the Gaussian. For several equally dominant saddles, add their contributions. Therefore, for the stated orientation,

$$I(s) \sim g(z_0)e^{sf(z_0)} i e^{-i\theta/2} \sqrt{\frac{2\pi}{s|f''(z_0)|}}. \quad (1.9.4)$$

Real special cases: Laplace method and stationary phase

The same local-Gaussian idea appears in two real-variable forms used later in the course.

Laplace's method (real, decaying exponential). For a smooth real phase φ with a unique interior maximum at x_0 , $\varphi'(x_0) = 0$, $\varphi''(x_0) < 0$, and a smooth amplitude g with $g(x_0) \neq 0$,

$$\int_a^b g(x) e^{s\varphi(x)} dx \sim g(x_0) e^{s\varphi(x_0)} \sqrt{\frac{2\pi}{s|\varphi''(x_0)|}}, \quad s \rightarrow +\infty. \quad (1.9.5)$$

This is (1.9.4) restricted to the real line with $\theta = \pi$ (so $f''(x_0) = -|\varphi''(x_0)|$ is already negative real and $\alpha = 1$). Chapter 2 uses this pattern for Stirling's approximation of $n!$ from the gamma integral.

Stationary phase (real, oscillatory). For a real phase φ on $[a, b]$ with $\varphi'(x_0) = 0$, $\varphi''(x_0) \neq 0$,

$$\int_a^b g(x) e^{is\varphi(x)} dx \sim g(x_0) e^{is\varphi(x_0)} e^{i(\pi/4) \operatorname{sgn} \varphi''(x_0)} \sqrt{\frac{2\pi}{s|\varphi''(x_0)|}}, \quad s \rightarrow +\infty, \quad (1.9.6)$$

Real special cases: Laplace method and stationary phase (cont.)

Here $\operatorname{sgn} q$ is $+1$ for $q > 0$ and -1 for $q < 0$. Assume the smooth amplitude is zero near the endpoints, x_0 is the only stationary point where the amplitude is nonzero, and $g(x_0) \neq 0$. Contributions from several nondegenerate stationary points are added. The Fresnel phase factor arises from the local Gaussian limit. Near x_0 ,

$$\varphi(x) \approx \varphi(x_0) + \frac{1}{2}\varphi''(x_0)(x - x_0)^2.$$

Set $u = \sqrt{s}(x - x_0)$, so $dx = du/\sqrt{s}$. Then

$$\int_a^b g(x) e^{is\varphi(x)} dx \sim e^{is\varphi(x_0)} \frac{g(x_0)}{\sqrt{s}} \int_{-\infty}^{\infty} e^{i(\varphi''(x_0)/2)u^2} du.$$

The oscillatory Gaussian has the corresponding formula

$$\int_{-\infty}^{\infty} e^{\pm iau^2} du = \sqrt{\frac{\pi}{a}} e^{\pm i\pi/4}, \quad a > 0.$$

We use the resulting Fresnel formula without its convergence proof. The positive and negative curvatures give the factors $e^{i\pi/4}$ and $e^{-i\pi/4}$, respectively: together they give (1.9.6). Chapter 5 applies this rule to the unit-circle integral for $J_n(x)$.

Real special cases: Laplace method and stationary phase (cont.)

Two refinements to carry forward. When the dominant point is an *endpoint* rather than an interior saddle or stationary point, repeated integration by parts replaces the Gaussian. The simplest case is *Watson's lemma*: for smooth g with sufficient decay,

$$\int_0^\infty g(t)e^{-st} dt \sim \frac{g(0)}{s} + \frac{g'(0)}{s^2} + \cdots, \quad s \rightarrow +\infty.$$

The endpoint supplies algebraic, not $1/\sqrt{s}$, decay.

Second, (1.9.4) assumes the amplitude g varies slowly near the saddle. If a pole of the integrand approaches or crosses the saddle—the geometry behind a transition region in wave problems—the single-Gaussian answer is no longer uniform, and an error-function-type transition function takes its place. Recognizing which refinement a problem needs is part of using the method.

Worked example: large binomial coefficient

Let m, n be integers with $0 < m < n$. As $n \rightarrow \infty$, keep $t = m/n$ away from 0 and 1. From $(1+z)^n = \sum_{k=0}^n \binom{n}{k} z^k$, Cauchy's coefficient formula gives

$$\binom{n}{m} = \frac{1}{2\pi i} \oint \frac{(1+z)^n}{z^{m+1}} dz = \oint g(z) e^{nf(z)} dz,$$

where

$$g(z) = \frac{1}{2\pi i z}, \quad f(z) = \log(1+z) - t \log z$$

(with branches chosen locally near the positive saddle below).

Saddle. Differentiate:

$$f'(z) = \frac{1}{1+z} - \frac{t}{z}, \quad f''(z) = -\frac{1}{(1+z)^2} + \frac{t}{z^2}.$$

Set $f'(z_0) = 0$:

$$z_0 - t(1+z_0) = 0 \implies z_0 = \frac{t}{1-t} = \frac{m}{n-m} > 0.$$

Worked example: large binomial coefficient (cont.)

Then $1 + z_0 = 1/(1 - t)$, and

$$\begin{aligned}f(z_0) &= \log \frac{1}{1-t} - t \log \frac{t}{1-t} = -t \log t - (1-t) \log(1-t), \\f''(z_0) &= -\frac{1}{(1+z_0)^2} + \frac{t}{z_0^2} = -(1-t)^2 + \frac{(1-t)^2}{t} \\&= (1-t)^2 \left(\frac{1}{t} - 1 \right) = \frac{(1-t)^3}{t} > 0.\end{aligned}$$

Thus $\theta = \arg f''(z_0) = 0$, so the descent direction from (1.9.3) is $\phi = \pi/2$: the local descent line is vertical. Take the contour in Cauchy's coefficient formula to be the circle $|z| = z_0$. On this circle, $|z|$ is fixed and

$$|1 + z_0 e^{i\psi}|^2 = 1 + 2z_0 \cos \psi + z_0^2$$

is largest at $\psi = 0$. The saddle $z = z_0$ therefore dominates. The counterclockwise tangent there points upward ($+i$), matching the descent direction and fixing the sign of the result.

Apply (1.9.4) with $s = n$.

$$\binom{n}{m} \sim \frac{1}{2\pi i z_0} e^{nf(z_0)} i \sqrt{\frac{2\pi}{nf''(z_0)}} = \frac{1}{2\pi z_0} e^{nf(z_0)} \sqrt{\frac{2\pi}{nf''(z_0)}}.$$

Worked example: large binomial coefficient (cont.)

Substitute $1/z_0 = (1-t)/t$ and $1/f''(z_0) = t/(1-t)^3$:

$$\frac{1}{2\pi z_0} \sqrt{\frac{2\pi}{nf''(z_0)}} = \frac{1-t}{2\pi t} \sqrt{\frac{2\pi t}{n(1-t)^3}} = \frac{1}{\sqrt{2\pi nt(1-t)}}.$$

Also $e^{nf(z_0)} = \exp[-n(t \log t + (1-t) \log(1-t))]$, so

$$\binom{n}{m} \sim \frac{\exp[-n(t \log t + (1-t) \log(1-t))]}{\sqrt{2\pi nt(1-t)}}. \quad (1.9.7)$$

Chapter 2's Stirling formula gives the same leading approximation from the factorial expression for $\binom{n}{m}$.

Summary

- Complex differentiability requires one direction-independent difference-quotient limit. With continuous first partial derivatives, this is equivalent to the Cauchy–Riemann equations $u_x = v_y$, $u_y = -v_x$.
- Cauchy's theorem implies path independence on simply connected domains. Cauchy's integral formula yields all derivatives and the local Taylor expansion.
- A Laurent series is valid on an annulus. Its principal part classifies an isolated singularity as removable, a pole, or essential. Branch points are not isolated and do not possess ordinary residues.
- The residue is the coefficient a_{-1} . The residue theorem evaluates a closed contour integral as $2\pi i$ times the sum of enclosed residues.
- Three real-integral templates: semicircle (rational/exponential), keyhole (powers and branch cuts, as in $\int_0^\infty x^a/(1+x)^2 dx$), and Jordan's indented semicircle (oscillatory integrals such as $\int \sin x/x dx$). Always check orientation and vanishing arcs before taking limits.
- **Steepest descent** (1.9.4) is a local Gaussian approximation at a nondegenerate dominant saddle. Real special cases: Laplace method (1.9.5) (Stirling in Ch. 2) and stationary phase (1.9.6) (large- x Bessel in Ch. 5). Orientation and the phase of the square root are part of the answer.